

Lab Protocol

Direct Immunoassay

Version: 3.0

Date: December 19, 2025

Author: Dakyung Kang

Lab: Biomedical Micro-Nano Systems Lab

Institution: University of Utah

Last Updated: December 19, 2025

Contents

1	Overview	3
1.1	Purpose	3
1.2	Scope	3
1.3	Expected Time	3
2	Safety Information	3
2.1	Personal Protective Equipment (PPE)	3
2.2	Chemical Hazards	3
2.3	Waste Disposal	3
3	Materials and Reagents	4
3.1	Equipment	4
3.2	Reagents	4
3.3	Solutions	5
4	Preparation	5
4.1	Before Starting	5
4.2	Solution Preparation	5
4.3	Target Window Preparation	6
4.4	Microfluidic Channel Preparation	6
5	Procedure	7
5.1	Step 1: [Cleaning the surface and PDMS sheet]	7
5.2	Step 2: [Bonding Patterned PDMS and Glass]	7
5.3	Step 3: [APTES Coating in the Pattern]	8
5.4	Step 4: [Direct Antigen Immobilization on APTES Surface with EDC/NHS Activation]	9
5.5	Step 5: [Washing and Blocking with BSA]	11
5.6	Step 6: [Attaching PDMS Microfluidic Channel Layer to the glass]	11
5.7	Step 7: [Preparing the Microbead]	12
5.8	Step 8: [Preparation of biotinylated detection antibody-streptavidin bead complex]	12
5.9	Step 9: [Introducing microbeads into the PDMS microfluidic Channel]	14
5.10	Step 10: [Washing the PDMS Microfluidic Channel]	14
6	Direct Immunoassay on CMOS	14
6.1	Step 1: [Cleaning the surface and PDMS sheet]	15
6.2	Step 2: [Bonding Patterned PDMS and Glass]	15
6.3	Step 3: [APTES Coating in the Pattern]	15
6.4	Step 4: [Direct Antigen Immobilization on APTES Surface with EDC/NHS Activation]	15
6.5	Step 5: [Washing and Blocking with BSA]	16
6.6	Step 6: [Preparing the Microbead]	16
6.7	Step 7: [Preparation of Biotinylated Detection Antibody-streptavidin Bead Complex]	16
6.8	Step 8: [Introducing Microbeads]	16
6.9	Step 9: [Washing the Surface]	16

7	Expected Results	17
7.1	Expected Observations	17
7.2	Typical Yield and Efficiency	17
7.3	Quality Control Criteria	17
8	Data Analysis	19
8.1	Calculations	19
8.2	Quality Control	20

1 Overview

1.1 Purpose

To establish and validate a direct immunoassay protocol on glass surfaces for electrode-based biosensor development. This protocol enables stepwise optimization of antigen immobilization, antibody-antigen binding, and microbead-based detection before final electrode integration.

1.2 Scope

This protocol applies to:

- Biosensor development using impedance-based electrode platforms
- Surface chemistry optimization on glass, silicon dioxide wafer, and aluminum-coated substrates
- Microfluidic immunoassay validation
- Human IgG detection in the range of 10 ng/mL to 100 ug/mL
- Magnetic microbead-based signal amplification strategies

1.3 Expected Time

- Preparation: 40 minutes
- Execution: 5 hours
- Analysis: 1 hour

2 Safety Information

CAUTION: This protocol involves chemical hazards, high temperatures, and UV exposure. Follow all safety guidelines and institutional protocols.

2.1 Personal Protective Equipment (PPE)

- Lab coat
- Safety goggles (required for plasma treatment and chemical handling)
- Nitrile gloves (recommended for chemical handling)
- Heat-resistant gloves (for oven and hotplate operations at 65-110°C)
- Face shield (recommended during oxygen plasma treatment)

2.2 Chemical Hazards

2.3 Waste Disposal

- **Chemical waste:** Collect acetone, IPA, and APTES waste in designated solvent waste containers
- **Biological waste:** Dispose of antibodies, BSA, and microbeads in biohazard waste
- **Sharps:** Dispose of biopsy punches and pipette tips in sharps containers
- **PDMS waste:** Cured PDMS can be disposed as regular solid waste; uncured should be treated as chemical waste

Chemical	Hazard	Precaution
Acetone	Flammable, irritant	Use in fume hood, avoid flames
APTES	Corrosive, moisture sensitive	Use in fume hood, handle with care
EDC/NHS	Irritant	Avoid skin/eye contact
Tween® 20	Mild irritant	Wear gloves
IPA	Flammable, irritant	Use in fume hood, avoid flames
PDMS (uncured)	Skin sensitizer	Wear gloves, avoid prolonged contact

Table 1: Chemical hazards and safety precautions

3 Materials and Reagents

3.1 Equipment

Fabrication Equipment

- Roland Camm-1 SERVO (cutter)
- Scotch tape
- Tweezers
- 2 mm biopsy punch

Lab Equipment

- Centrifuge
- Vortex mixer
- Roller mixer (shaking mode)
- 96-well plate with magnets
- N₂ gas supply

Consumables

- Petri dishes (100mm × 15mm)
- Microtubes
- Plastic cups
- Various pipette tips
- Kimtech paper

3.2 Reagents

Note: Following section contains an embedded hyperlink

- **APTES:** 3-Aminopropyltriethoxysilane (Cat# 440140, sigma-aldrich)
- **EDC:** 1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide (Cat# 22980, thermo scientific)
- **NHS:** N-Hydroxysuccinimide (Cat# 24500, thermo scientific)
- **BSA:** Bovine Serum Albumin (Cat# A7906, sigma-aldrich)

- **Tween® 20:** Polysorbate 20 (Cat# 1002942331, sigma-aldrich)
- **Human IgG:** 4 mg/mL in Saline Solution
- **Biotinylated Anti-Human IgG:** Goat Anti-Human IgG Biotin
- **Dynabeads™ M-280 Streptavidin:** 10 mg/mL (Cat# 11205D, thermo scientific)
- **PDMS:** Polydimethylsiloxane with curing agent (10:1 ratio)
- **Acetone, IPA:** For cleaning

3.3 Solutions

- **Buffer A:** PBS 1× (pH 7.4)
- **Buffer B:** MES 100 mM (pH 5.8-6.5)

4 Preparation

4.1 Before Starting

1. Check that all equipment is available and functional
2. Prepare all required solutions
3. Set up workspace and equipment

4.2 Solution Preparation

Solution 1 (PBS T 0.1% to 1%):

1. Prepare Tween® 20 solution and 1X PBS
2. To make 10% PBS T solution, mix 100 uL of Tween® 20 + 900 uL of 1x PBS
3. To make 1% PBS T, mix 100 uL of 10% PBS T + 900 uL of 1x PBS
4. To make 0.1% PBS T, mix 100 uL of 1% PBS T + 900 uL of 1x PBS

Solution 2 (BSA 0.1% to 1%):

1. Prepare Bovine Serum Albumin and 1X PBS
2. To make 1% BSA, mix 10mg of BSA and 1,000 uL of 1X PBS
3. To make 0.1% BSA, mix 100 uL of 1% BSA and 900 uL of 1X PBS

Solution 3 (EDC):

1. Prepare 5µL of EDC (Ratio 0.4mg of EDC/ 50µL of 100 mM MES)

Solution 4 (NHS):

1. Prepare 5µL NHS (Ratio 1.1mg of NHS/50µL of 100 mM MES)

Solution 5 (3% APTES):

1. Add 50 uL of APTES and 1,617 uL of DI Water in a microtube

4.3 Target Window Preparation

Printing PDMS Target Window Sheet

1. Power on Roland Camm-1 SERVO and provided laptop
2. Cut approximately 22-24 inches lengthwise of the sticker sheet
3. Adjust rollers to appropriate positions, load the sticker sheet, and secure with the back lever. Rollers must be positioned in the right regions
4. Press the down arrow on the Roland Camm-1 to select "piece", then press enter to collect sheet dimensions
5. Open the Auto CAD drawing file and export it as DXF
6. Import and open the file in Illustrator
7. Select the area that you want to print
8. Go to "Window", select "Extension", and "Roland Cut Studio"
9. Click the "Model GX-24", then "Cut"
10. Keep the PDMS sheet in the petri dish to avoid the dust

4.4 Microfluidic Channel Preparation

Printing Stickers for PDMS Mold

See Section 4.3 for details.

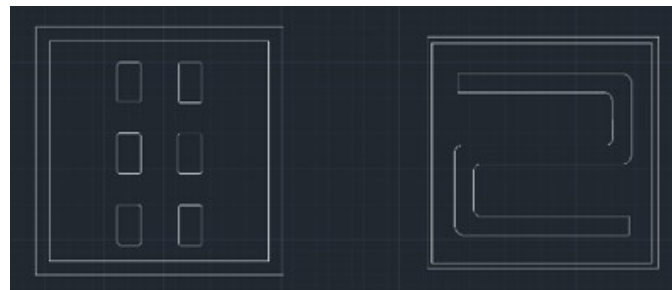


Figure 1: AutoCAD Design: PDMS Sheet and Mold

Creating the Petri Molds

1. Prepare a 100mm x 15mm size of a petri dish, scotch tape, and the cutout of your sticker
2. Place the sticker sheet cut side up and layer with scotch tape, try and keep tape as flat as possible
3. Carefully remove the tape from the table with the sheet still attached
4. Place the sheet tape side down and peel the bottom layer off, exposing the silver colored sticky layer
5. The bottom layer will be applied by pressing the Petri **lid** on the center of the sticker
6. Carefully remove the tape, then with tweezers, remove the sticker that surrounds the channels

Making the PDMS Solution

1. Put the plastic cup on a scale and zero it, then combine 25g of PDMS with 2.5g of curing agent (10:1 ratio)
2. Stir for 10 minutes, until all air bubbles are removed
3. Place the mixture in the degasser vacuum and cover securely with the plastic lid
4. Close the exhaust valve by rolling the blue roller back
5. Turn on degasser vacuum beneath the table until the solution bubbles up to 90% cup height
6. Turn off the vacuum and monitor the solution periodically to avoid overfilling
7. Wait until all bubbles have popped (5-10 minutes) before opening the valve
8. Pour the PDMS solution into the petri dish
Note: Do NOT pour over the center or sticker, add along the sides
9. Repeat steps 3 to 7

Preparation and Curing of PDMS Molds

1. Place the petri dish on the hotplate set to 65°C, leave it completely cured (2 hours) or overnight
Note: Or cure the mold in room temperature for 2 days

5 Procedure

5.1 Step 1: [Cleaning the surface and PDMS sheet]

1. Use acetone, IPA, and DI water to clean the glass surface
2. Dry the glass surface the N₂ gas
3. Use IPA and DI water to clean the PDMS surface
4. Dry the PDMS surface the N₂ gas

5.2 Step 2: [Bonding Patterned PDMS and Glass]

1. Under the fume hood, attach the PDMS sheet to the surface
2. Place the glass into a 100mm x 15mm size of petri dish
3. Place the petri dish in the oven (110°C) for 15 minutes
4. Take out the petri dish from the oven

5.3 Step 3: [APTES Coating in the Pattern]

1. Oxygen Plasma PDMS surface with glass (OH-Glass)
2. Transfer the glass to the petri dish
3. Drop 2uL of 3% APTES per window twice
4. Wait until the chemical reaction for 40 minutes at room temperature
Note: Wet Kimtech paper with DI water must be placed in the petri dish
5. Prepare 2 petri dishes with DI water to dip/wash to remove excessive APTES solution
6. After 40 minutes, dip the APTES-bonded glass gently into the first Petri dish
7. Take out the APTES bonded glass gently from the first Petri dish and dip it into the second Petri dish
8. After taking out the APTES-bonded glass from the second Petri dish, place it into the oven (110°C) for 30 minutes

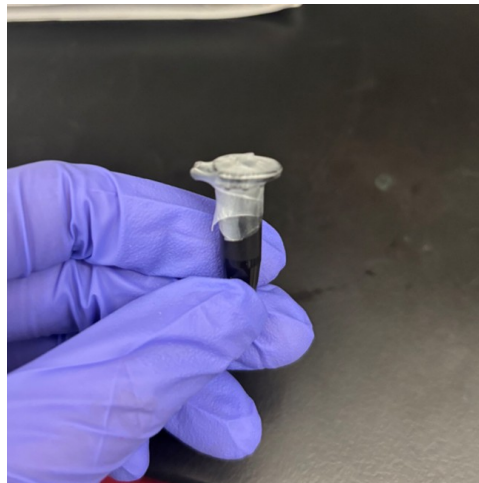


Figure 2: Aliquote from APTES Bottle

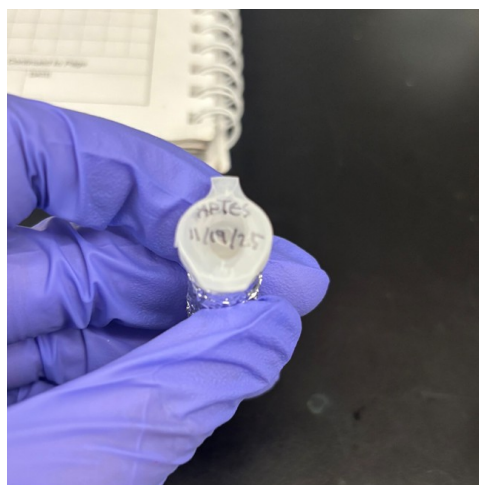


Figure 3: 3% of APTES

5.4 Step 4: [Direct Antigen Immobilization on APTES Surface with EDC/NHS Activation]

1. After taking out the APTES-bonded glass from the oven, place the APTES-bonded glass in room temperature for 10 minutes
2. Take out the list of chemicals from the -75°C freezer in the bio lab
 - 1 vial (5 uL) of 0.4mg of EDC/50μL of 100 mM MES(Figure 4)
 - 1 vial (5 uL) of 1.1mg of NHS/50μL of 100 mM MES(Figure 5)
 - 1 vial (10 uL) of 4 mg/mL Human IgG in Saline Solution(Figure 6)

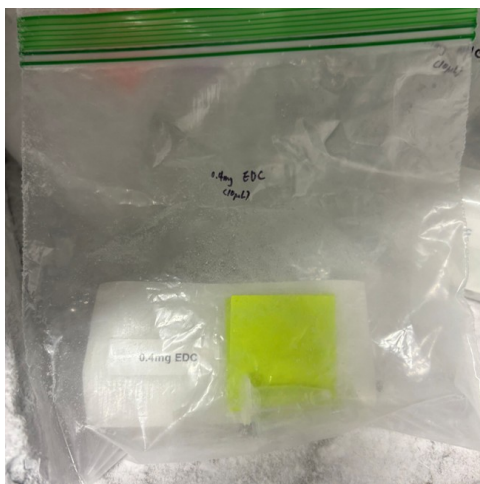


Figure 4: 0.4mg of EDC/50μL of 100 mM MES

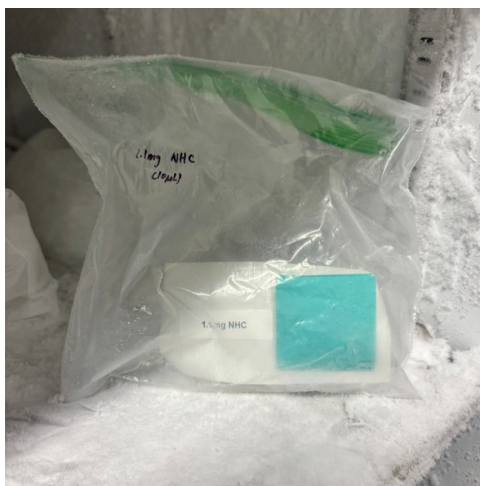


Figure 5: 1.1mg of NHS/50μL of 100 mM MES

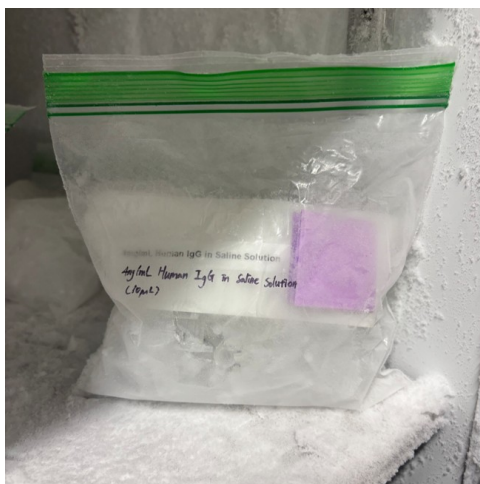


Figure 6: 4 mg/mL Human IgG in Saline Solution

3. Add 5 uL of 1x PBS to the vial of 5 uL of 4 mg/mL Human IgG in Saline Solution
4. Add 5 uL of 0.4 mg EDC into the Human IgG vial
5. Add 5 uL of 1.1 mg NHS into the Human IgG vial
6. A total of 20 uL of 1000 ug/mL Human IgG ($20 \text{ ug Human IgG} / 20 \text{ uL} = 1 \text{ mg/mL} = 1000 \text{ ug/mL}$)
7. Wait until the chemical reaction is finished at room temperature for 15 minutes
8. Dilute to the desired concentration
 - Target concentration: 100ug/mL, 10ug/mL, 1ug/mL, 100ng/mL, 10ng/mL
 - 90uL of 1X PBS + 10uL of 1000ug/mL Human IgG (total 100uL of 100ug/mL)
 - 90uL of 1X PBS + 10uL of 100ug/mL Human IgG (total 100uL of 10ug/mL)
 - 90uL of 1X PBS + 10uL of 10ug/mL Human IgG (total 100uL of 1ug/mL)
 - 90 uL of 1X PBS + 10uL of 1ug/mL Human IgG (total 100uL of 100ng/mL)
 - 90 uL of 1X PBS + 10uL of 100ng/mL Human IgG (total 100uL of 10ng/mL)
9. Transfer the APTES-bonded glass to a Petri dish with wet Kimtech paper with DI Water
10. Use a gel loading pipette tip and fill the window with Human IgG(Figure 7)

Window 1: Blank	Window 2: 100 ug/mL
Window 3: 1 ug/mL	Window 4: 10 ug/mL
Window 5: 100 ng/mL	Window 6: 10 ng/mL
11. Wait until the chemical reaction is finished at room temperature for 1 hour

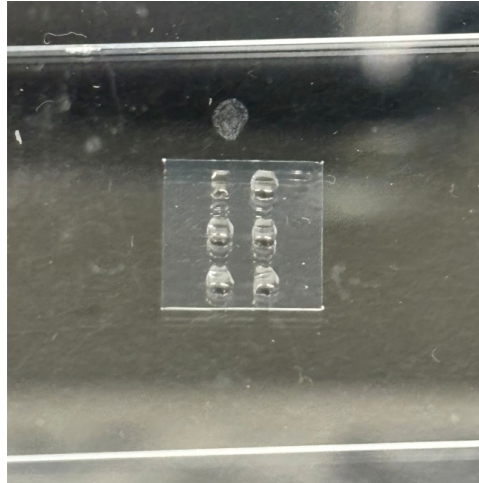


Figure 7: After Introducing Each Concentration of Solution

5.5 Step 5: [Washing and Blocking with BSA]

1. Use a gel loading pipette tip and carefully remove the Human IgG from the window
2. Add 0.1% BSA into the window just enough to fill the window to remove excessive Human IgG
3. Repeat 1-2 steps
4. Remove the patterned PDMS sheet carefully without disturbing the Human IgG-coated window
5. Close the lid of the Petri dish to avoid any dust particles

5.6 Step 6: [Attaching PDMS Microfluidic Channel Layer to the glass]

1. Take out the PDMS microfluidic channel layer from the Petri Dish
2. Use a 2 mm biopsy punch to punch the inlet and outlet
3. Use IPA and DI Water to clean the surface of the PDMS layer
4. Dry the glass surface the N_2 gas
5. Oxygen plasma treatment on the PDMS layer only, not on the glass
6. Align the PDMS layer based on the Human IgG pattern on the glass and bond together (Figure 8)
7. Wait for 10-15 minutes to bond
8. Passivate the microfluidic channel by adding 5 μ L of 1% BSA into the inlet
9. In room temperature, wait for 10 minutes to minimize non-specific binding

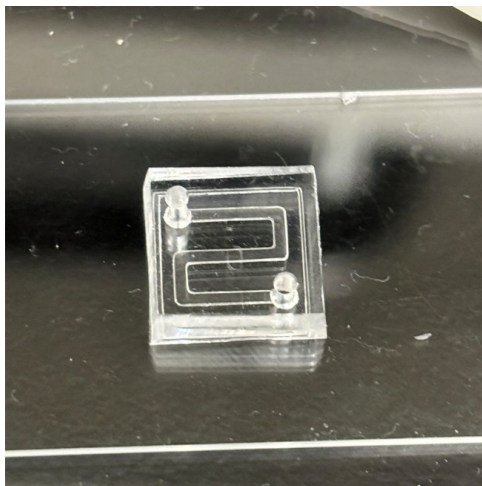


Figure 8: After Attaching PDMS Microfluidic Channel

5.7 Step 7: [Preparing the Microbead]

Microbead (Dynabeads™ M-280 Streptavidin 10mg/mL) Preparation

1. Rotate the vial on the table with one hand gently to mix the microbead
2. Use a gel loading pipette tip to transfer 15 uL from the microbead vial to the microtube
3. Add 85 uL of 0.1% PBS T in the microtube (Total 100 uL in the microtube)
4. Use Vortex to wash the microbead with 0.1% PBS T for 5 s
5. Place the microtube in the centrifuge and spin down the microbead to the end of the microtube for 10-15 s
6. Place the microtube into a 96-well with a magnet to hold the microbead for 1 minute
7. Use a gel loading pipette tip and remove the liquid only
8. Add 100 uL of 0.1% PBS T in the microtube
9. Repeat steps 4-8 3 times
10. Add 15 uL of 0.1% BSA into the microtube (15uL 0.1% BSA + 15uL microbead)
11. Store the microbead in the fridge

5.8 Step 8: [Preparation of biotinylated detection antibody-streptavidin bead complex]

1. Take out and prepare 10 uL of 0.6 mg/mL Goat α -H (Anti-Human) IgG Biotin from the -75°C freezer in the bio lab (Figure 9)
2. Transfer 5 uL of 0.6 mg/mL Goat α -H (Anti-Human) IgG Biotin into the new microtube
3. Add 245 uL of 0.1% BSA into 5 uL of biotinylated anti-human IgG microtube
4. Take out the microbead with 15 uL of 0.1% BSA from the fridge
5. Add 15 uL of the microbead with 0.1% BSA into the biotinylated

6. Place the microtube into the roller and mix with shaking mode for 30 minutes at room temperature (Figure 10)
7. Take out the microtube from the roller, place the microtube in the centrifuge, and spin down the microbead to the end of the microtube for 10-15 seconds
8. Place the microtube into 96 96-well plate with magnets to hold the microbead for 1 minute
9. Use a gel loading pipette tip and remove the liquid only
10. Add 100 uL of 1% PBS T in the microtube
11. Use Vortex to wash the microbead
12. Repeat step 7-12 3 times
13. Add 15 uL of 0.1% BSA into the microtube (15 uL 0.1% BSA + 15 uL microbead)
14. Store the microbead with the biotinylated detection antibody in the fridge

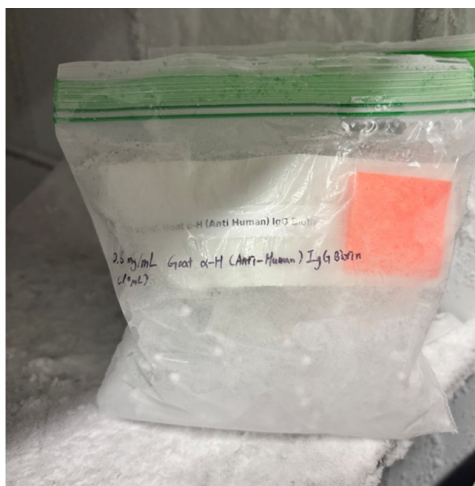


Figure 9: 0.6 mg/mL Goat α -H (Anti-Human) IgG Biotin

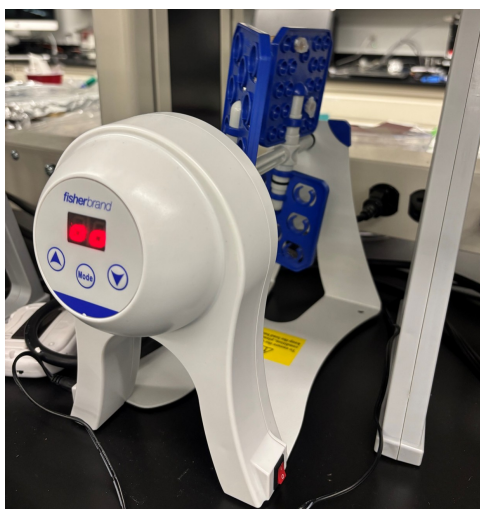


Figure 10: Reactive Goat α -H (Anti-Human) IgG Biotin with Dynabead

5.9 Step 9: [Introducing microbeads into the PDMS microfluidic Channel]

1. After passivating the microfluidic channel with 1% BSA, set where to introduce the microbeads with different concentrations
2. Prepare and trim the filtered paper to apply vacuum/negative pressure from the inlet to the outlet
3. Add 5 μL of Detection Ab-Bead into the inlet
4. Place the filtered paper on the outlet and apply vacuum/negative pressure
5. Wait until the chemical reaction is finished in room temperature for 30 minutes

CAUTION: Air bubbles must not be seen in the microfluidic channel

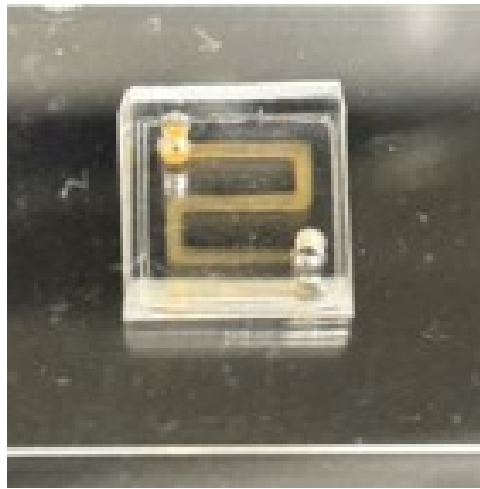


Figure 11: After Introducing the Microbead in the Channel

5.10 Step 10: [Washing the PDMS Microfluidic Channel]

1. Prepare and trim a 1 mL pipette tip to wash the microfluidic channel
2. Using $p = p_0 + \rho gh$, calculate the pressure based on the height of the pipette tip
3. After 30 minutes of the chemical reaction, place the trimmed pipette tip into the inlet gently
4. Using a gel loading pipette tip, add 0.1% PBST into the trimmed pipette tip based on the height
5. Remove the excessive microbeads and 0.1% PBST solution from the outlet
6. Repeat highlighted steps 3 times
7. Washing again with 1X PBS
8. Remove the trimmed pipette tip gently
9. Cover the inlet and the outlet with the PDMS sheet to avoid evaporation

6 Direct Immunoassay on CMOS

The direct Immunoassay on the CMOS chip has a similar way to the test on the silicon dioxide wafer.

6.1 Step 1: [Cleaning the surface and PDMS sheet]

Use the below PDMS pattern (Figure 12).

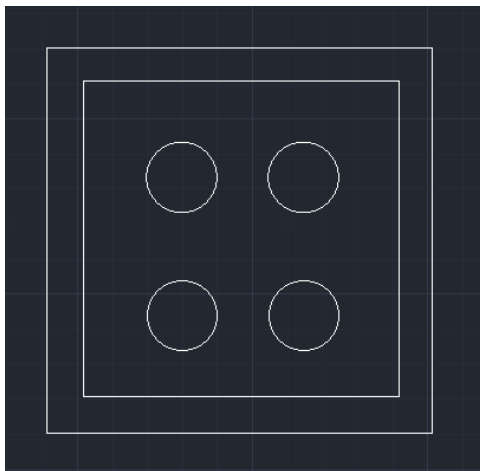


Figure 12: PDMS Pattern for Direct Immunoassay on the CMOS Chip

See Section 5.1 for details.

6.2 Step 2: [Bonding Patterned PDMS and Glass]

See Section 5.2 for details.

6.3 Step 3: [APTES Coating in the Pattern]

Cover the entire PCB Part with Kapton tape (polyimide tape) as shown in Fig. 13.

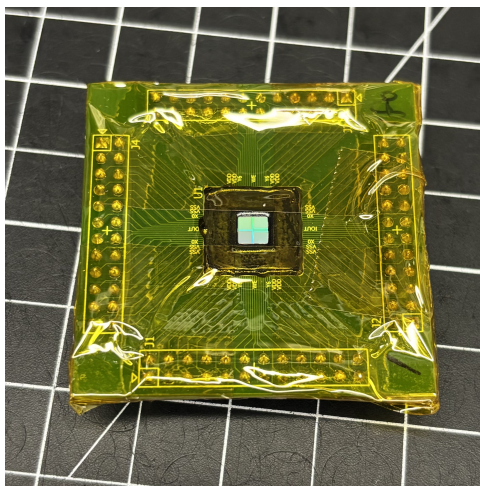


Figure 13: Covered Chip with Kapton Tape

See Section 5.3 for details.

6.4 Step 4: [Direct Antigen Immobilization on APTES Surface with EDC/NHS Activation]

Prepare the target concentrations: 100 ug/mL, 1 ug/mL, and 100 ng/mL, as shown in Section 5.4.

6.5 Step 5: [Washing and Blocking with BSA]

See Section 5.5 for details.

6.6 Step 6: [Preparing the Microbead]

See Section 5.7 for details.

6.7 Step 7: [Preparation of Biotinylated Detection Antibody-streptavidin Bead Complex]

See Section 5.8 for details.

6.8 Step 8: [Introducing Microbeads]

1. Passivate the entire surface with 1% BSA
2. Drop 4uL of Dynabeads M280-Streptavidin for each window
3. Wait until the chemical reaction is finished at room temperature for 30 minutes

6.9 Step 9: [Washing the Surface]

1. After 30 minutes of the chemical reaction, carefully drop the 0.1% PBS T onto the chip surface
2. Place the chip upside down, and wait until the 0.1% PBS T is going to fall (Figure 14)
3. Using Kimtech tissue to absorb the remaining 0.1% PBS T solution (Figure 15)
4. Repeat step 2-3 3 times



Figure 14: Chip inversion for 0.1% PBS T drainage

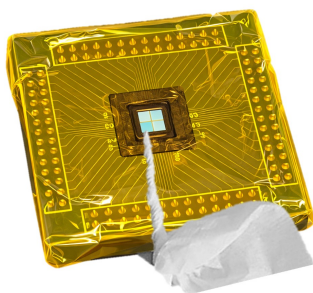


Figure 15: Absorption of Residual PBS T with Kimtech tissue

7 Expected Results

7.1 Expected Observations

- **APTES Coating:** Hydrophobic surface after APTES treatment; water droplets should bead up on the surface
- **Antigen Immobilization:** Uniform filling of microfluidic windows without air bubbles; clear concentration gradient visible across windows (Blank, 100 ug/mL, 10 ug/mL, 1 ug/mL, 100 ng/mL, 10 ng/mL)
- **PDMS Bonding:** Strong, irreversible bond between PDMS layer and glass after oxygen plasma treatment; no leakage during fluid introduction
- **Microbead Detection:** Concentration-dependent microbead binding pattern; higher antigen concentrations should show greater microbead accumulation
- **Negative Control (Blank):** Minimal to no microbead binding, confirming BSA blocking effectiveness
- **Visual Appearance:** Brown microbeads (Dynabeads M-280) clearly visible under microscope in positive samples

7.2 Typical Yield and Efficiency

- **PDMS Yield:** 25 g PDMS + 2.5 g curing agent produces sufficient material for multiple chips
- **Detection Sensitivity:** Linear response range from 10 ng/mL to 100 ug/mL Human IgG
- **Signal-to-Noise Ratio:** >3:1 for concentrations ≥ 100 ng/mL compared to blank
- **Reproducibility:** Coefficient of variation (CV) <15% across replicate windows

7.3 Quality Control Criteria

- **Surface Cleanliness:** No visible particles or contamination after N₂ drying
- **PDMS Curing:** Solid, non-sticky surface after 2 hours at 65°C
- **Channel Integrity:** No leakage during fluid flow; uniform channel dimensions

- **Negative Control:** Blank window shows <10% of signal compared to 100 ug/mL window
- **Positive Control:** 100 ug/mL window shows clear microbead accumulation
- **Dose Response:** Monotonic increase in microbead density with increasing antigen concentration

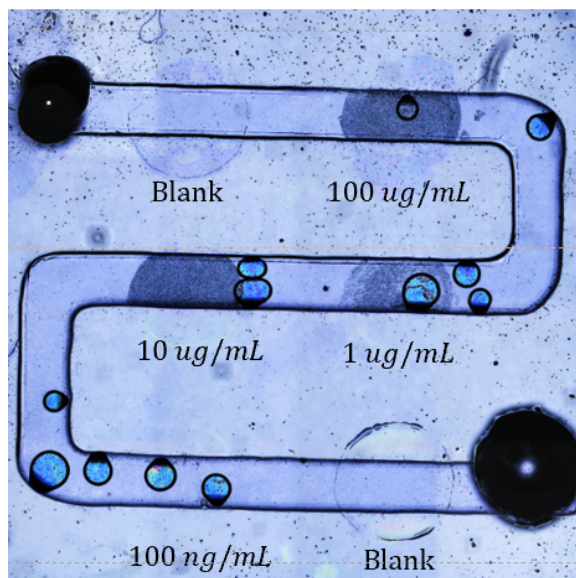


Figure 16: Direct Immunoassay: Circle Pattern on Silicon dioxide Wafer

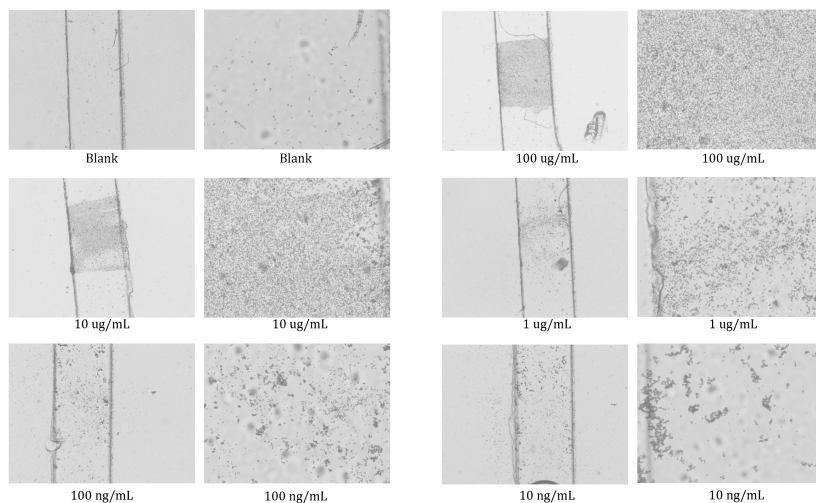


Figure 17: Direct Immunoassay: Rounded Rectangle Pattern on the Glass

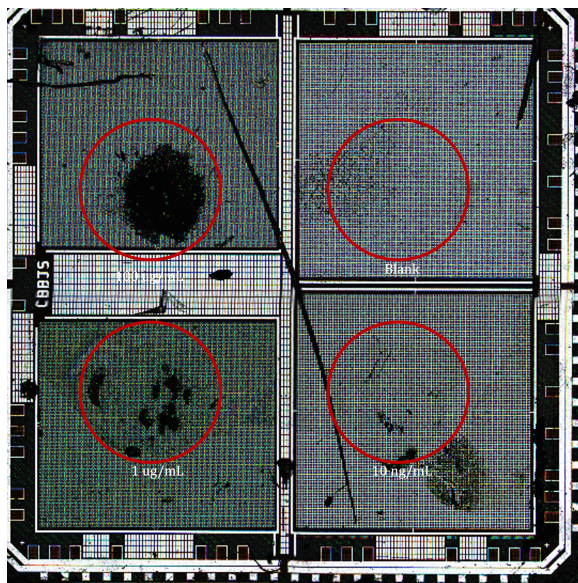


Figure 18: Direct Immunoassay: Circle Pattern on CMOS Chip

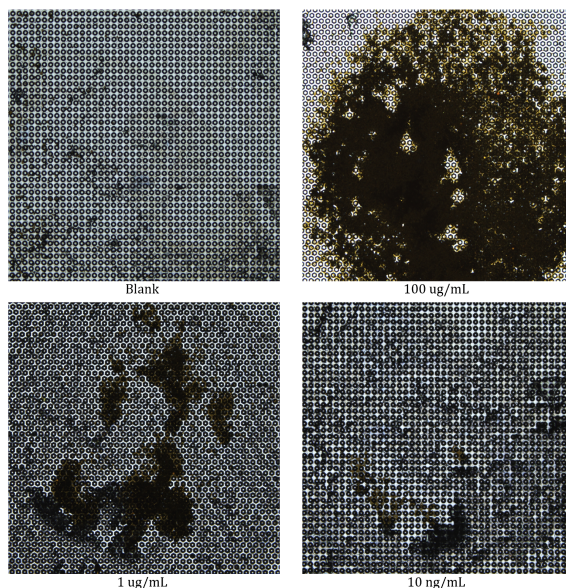


Figure 19: Zoom in View of CMOS Chip

8 Data Analysis

8.1 Calculations

Antigen Dilution Calculations:

$$C_1 V_1 = C_2 V_2$$

where C_1 = initial concentration, V_1 = initial volume, C_2 = final concentration, V_2 = final volume

Example - Preparing 10 ug/mL from 100 ug/mL:

$$(100 \text{ ug/mL})(10 \text{ uL}) = (10 \text{ ug/mL})(100 \text{ uL})$$

Therefore: 10 uL of 100 ug/mL + 90 uL of $1 \times$ PBS

Microbead Density Calculation:

$$\text{Bead Density} = \frac{\text{Number of Beads}}{\text{Window Area (mm}^2\text{)}}$$

Signal-to-Noise Ratio:

$$\text{SNR} = \frac{\text{Signal}_{\text{sample}}}{\text{Signal}_{\text{blank}}}$$

Pressure Calculation for Washing:

$$p = p_0 + \rho gh$$

where p = total pressure, p_0 = atmospheric pressure, ρ = fluid density (1000 kg/m³ for water), g = 9.8 m/s², h = height of fluid column

8.2 Quality Control**Pre-Experiment QC:**

- Verify all reagents are within expiration dates
- Confirm antibody and protein storage conditions (-75°C for frozen stocks)
- Check PDMS mixing ratio (10:1) accuracy using scale
- Inspect oxygen plasma functionality before surface treatment

During Experiment QC:

- Monitor incubation times precisely (± 2 minutes acceptable)
- Verify no air bubbles in microfluidic channels during bead introduction
- Check for proper PDMS-glass bonding (no delamination under gentle pressure)

Post-Experiment QC:

- **Negative Control (Blank):** Signal <10% of highest concentration
- **Positive Control (100 ug/mL):** Clear, reproducible microbead binding
- **Linearity:** Dose-response curve shows monotonic increase
- **Reproducibility:** Triplicate measurements show CV < 15%
- **No Cross-Contamination:** Adjacent windows show distinct concentration-dependent responses

Acceptance Criteria:

- All QC checks passed
- SNR ≥ 3 for concentrations ≥ 100 ng/mL
- Dose-response correlation coefficient $R^2 \geq 0.95$
- No visible contamination or channel defects
- Negative control validates blocking efficiency